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SECTION 2

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PROJECT PROPOSAL

TITLE:

**BUSINESS INTELLIGENCE ANALYTICS FOR U.S. E-COMMERCE PERFORMANCE
OPTIMIZATION: A MULTI-DIMENSIONAL ANALYSIS OF SALES, CUSTOMER
BEHAVIOUR, AND DELIVERY OPERATIONS (2019–2025)**

GROUP

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1.0 Title

Business Intelligence Analytics for U.S. E-Commerce Performance Optimization: A Multi-Dimensional Analysis of Sales, Customer Behaviour, and Delivery Operations (2019–2025)

2.0 Problem Background

2.1 Business context

The U.S. e-commerce industry has experienced dramatic growth over the past decade, accelerated by the pandemic and the rapid adoption of online shopping. By 2024, U.S. e-commerce sales exceeded USD 1.1 trillion. Online marketplaces now operate at massive scale, manage thousands of daily orders, millions of customer interactions and coordinate national logistics networks involving sellers, carriers, and end consumers. However, many e-commerce businesses struggle to transform raw transactional data into useful business insight. They face some operational challenges including understanding which product categories and regions drive the highest revenue, identifying the root causes of delivery delays, understanding how payment behaviour varies across customer segments, and determining what factors most strongly influence customer satisfaction scores. Without a structured Business Intelligence solution, company managers are forced to make critical decisions rather than data driven insights.

2.2 Importance of the problem

When e-commerce does not run smoothly, it loses money in several key ways. First, delivery delays directly increase customer leave and return rates. Poor product assortment decisions will result in wasted money on warehouse storage. Furthermore, without insight into seller performance and customer review, sellers cannot identify underperforming partners or address quality issues before they damage brand reputation. A Business Intelligence (BI) solution fixes this. By combining data on sales, customer habits, payments, and shipping locations, managers get a complete view of the business. Instead of guessing, they can use real data to make fast decisions that directly improve profitability, customer satisfaction, and logistics efficiency.

2.3 Expected impact

- Higher Profits: Finding our most profitable products and best customers will help us know exactly where to spend our marketing budget and what inventory to stock.
- Faster Shipping: By tracking exactly when and where delivery delays happen, the logistics team can fix shipping routes and choose better delivery partners.
- Happier Customers: Understanding how people pay and why they leave reviews will help us create better loyalty programs and improve the overall shopping experience.
- Better Sellers: By tracking how fast sellers ship and how much they charge, we can remove bad sellers and reward the good ones.
- Smarter Planning: Looking at sales trends from 2019 to 2025 will help the business predict future sales, budget for shipping costs, and prepare for busy holiday seasons.

3.0 Objective

The primary goal of this project is to transform raw e-commerce data into useful business insights.

- To integrate and clean multiple data sources: To combine separate datasets into one unified, clean database using Alteryx.
- To analyse sales performance: To analyse overall sales revenue trends, product category performance, and regional distribution of orders across the United States from 2019 to 2025 to identify high growth segments and underperforming areas.
- To analyze product profitability: To calculate the actual profit margins of different product categories by subtracting manufacturing and shipping costs from the final selling price.
- To evaluate logistics and delivery performance: To track delivery delays and shipping costs across different states to identify bottlenecks in the supply chain.
- To understand customer satisfaction: To explore the correlation between review scores, delivery delay, discount rates, and product categories, and to derive actionable recommendations to improve the overall customer experience on the platform.

4.0 Datasets Description

4.1 Source of each dataset

The primary data source for this project is the Synthetic U.S. E-Commerce Dataset (2019–2025, 1 Million Orders) obtained from Kaggle (<https://www.kaggle.com/datasets/limjeongeun/synthetic-u-s-e-commerce-dataset-1m-orders>). This comprehensive dataset simulates realistic e-commerce transactions and operations across the United States over a six-year period.

The dataset consists of multiple relational CSV files:

1. Orders Dataset – Contains transactional records of customer purchases
2. Products Dataset – Includes product catalog information and pricing
3. Customers Dataset – Stores customer demographic and location data
4. Payments Dataset – Records payment methods and transaction details
5. Sellers Dataset – Contains seller information and performance metrics
6. Delivery Dataset – Tracks shipping and logistics information
7. Reviews Dataset – Captures customer feedback and satisfaction ratings

All datasets are synthetically generated to represent realistic U.S. e-commerce scenarios while ensuring data privacy and compliance. The synthetic nature allows for comprehensive analysis without confidentiality concerns.

4.2 Type of data (structured/unstructured)

All datasets in this project are structured data stored in CSV (Comma-Separated Values) format. Structured data refers to information organized in a tabular format with clearly defined rows and columns, where each column represents a specific attribute and each row represents a record.

Characteristics of the structured datasets:

- Relational structure: Datasets are linked through common keys (e.g., Order ID, Customer ID, Product ID, Seller ID)
- Predefined schema: Each table has a consistent set of columns with specific data types
- Quantitative and categorical variables: Includes numerical values (prices, quantities, dates) and categorical attributes (product categories, states, payment methods)
- Temporal data: Contains date and timestamp fields enabling time-series analysis

This structured format is ideal for Business Intelligence analysis as it allows for efficient querying, aggregation, joining operations, and visualization in BI tools like Alteryx and Tableau.

4.3 Key variables /features

The following tables outline the key variables from each dataset that will be utilized in the analysis:

Orders Dataset

Variable Name	Data Type	Description
Order ID	String	Unique identifier for each order
Customer ID	String	Links to customer information
Order Date	Date	Date when order was placed
Order Status	Categorical	Status of order (delivered, shipped, cancelled, etc.)
Order Value	Numeric	Total value of the order

Products Dataset

Variable Name	Data Type	Description
Product ID	String	Unique identifier for each order
Product Category	Categorical	Classification of product type
Product Name	String	Name/description of the product
Selling Price	Numeric	Final price paid by customer
Manufacturing Cost	Numeric	Cost to produce/procure the product
Original Price	Numeric	List price before discounts

Customers Dataset

Variable Name	Data Type	Description
Customer ID	String	Unique identifier for each customer
Customer State	Categorical	U.S. state where customer is located
Customer City	String	City of customer residence
Customer Zip Code	String	Postal code for geographic analysis

Payments Dataset

Variable Name	Data Type	Description
Order ID	String	Links to order information
Payment Method	Categorical	Type of payment (credit card, debit, PayPal, etc.)
Payment Value	Numeric	Amount paid
Payment Date	Date	Date of payment transaction

Sellers Dataset

Variable Name	Data Type	Description
Sellers ID	String	Unique identifier for each seller
Seller State	Categorical	U.S. state where seller is located
Seller Performance Score	Numeric	Rating of seller reliability
Shipping Fee Charged	Numeric	Shipping cost charged by seller

Delivery Dataset

Variable Name	Data Type	Description
Order ID	String	Links to order information
Shipping Carrier	Categorical	Delivery service provider
Estimated Delivery Date	Date	Expected delivery date
Actual Delivery Date	Date	Real delivery date
Shipping Cost	Numeric	Cost of shipping and handling

Reviews Dataset

Variable Name	Data Type	Description
Review ID	String	Unique identifier for each review
Order ID	String	Links to order information
Review Score	Numeric	Customer rating (typically 1-5 scale)
Review Date	Date	Date when review was submitted
Review Comment	Text	Customer feedback (optional analysis)

5.0 Proposed BI Approach

5.1 BI Tools and Visualization

This project will primarily use Alteryx Designer for data preparation, transformation, and analytical workflow development. Alteryx provides a visual drag-and-drop interface that simplifies data integration from multiple datasets while enabling efficient cleaning and feature engineering. For dashboard development and data presentation, Tableau will be used to create interactive visualisations and business dashboard. Tableau allows stakeholders to explore insights dynamically through filters, trend analysis, and geographical mapping. The integration

between Alteryx and Tableau enables a complete Business Intelligence pipeline, where raw e-commerce data is transformed into actionable insights for decision-making.

5.2 Data Integration, Cleaning, and Transformation Method

The project dataset contains transactional records related to orders, products, customers, payments, sellers, and delivery operation from 2019 to 2025.

The data preparation process in Alteryx will involve the following steps:

Alteryx Tool	Purpose
Input Data	Import all raw CSV files
Select	Keep relevant attributes and remove unnecessary columns
Data Cleanse	Remove duplicates, null values, and inconsistent formatting
Filter	Exclude invalid records and outliers
Join	Merge datasets using common keys such as Order ID and Customer ID
Formula	Create derived analytical variables
Summarize	Aggregate KPI metrics
Output Data	Export processed datasets

Transformation Methods:

Several calculated fields will be created to support analysis:

1. Delivery Delay (Days)
 $\text{Actual Delivery Date} - \text{Estimated Delivery Date}$
2. Profit Margin (%)
 $(\text{Selling Price} - \text{Manufacturing Cost} - \text{Shipping Cost}) / \text{Selling Price} \times 100$
3. Discount Rate (%)
 $(\text{Original Price} - \text{Selling Price}) / \text{Original Price} \times 100$

5.3 Planned Analysis

The analysis will focus on five major business intelligence dimensions:

1. Sales Performance Analysis

Analyze revenue trends across years, product categories, and states to identify growth patterns.

2. Product Profitability Analysis

Evaluate which products and categories generate the highest margins.

3. Logistics and Delivery Performance Analysis

Identify delivery bottlenecks, delayed shipment regions, and shipping efficiency.

4. Customer Satisfaction Analysis

Examine the relationship between review scores, delivery delays, and discount strategies.

5. Customer Behavior Analysis

Study customer purchasing patterns, preferred payment methods, and order frequency.

5.4 Proposed Visualization Forms

The following visualization forms will be used in Tableau dashboards:

Visual Form	Purpose
Line Chart	Show revenue trends over time
Bar Chart	Compare product category and regional performance
Choropleth Map	Display geographic distribution of orders and delays
Scatter Plot	Analyse correlation between delivery delay and customer reviews
Heatmap	Identify seasonal and peak purchasing periods
Donut Chart	Show payment method distribution
Box Plot	Compare delivery delay variability across shipping methods

These visualizations are selected because they provide clear insight into sales trends, operational performance, and customer behaviour patterns.